Engineering Findings

1. Multi-objective behaviour of the extractive distillation process

The engineering analysis revealed that the extractive distillation process is governed by competing objectives. No single operating condition simultaneously maximized ethanol purity, ethanol recovery, ethylene glycol recovery, and minimized energy consumption. Improvements in one performance metric were generally accompanied by deterioration in another, demonstrating the necessity of multi-objective optimization.

The Process Performance Index (PPI) successfully balanced these conflicting objectives and identified operating conditions that achieved the best overall process performance.

2. Effect of Column 1 Boilup Ratio

The boilup ratio was identified as one of the most influential operating variables affecting overall process performance.

Major observations include:

- Maximum ethanol purity of **97.68 wt%** occurred at a boilup ratio of **0.44**.
- Increasing boilup beyond **0.44** steadily reduced ethanol purity to approximately **92 wt%**.
- Ethanol recovery continuously increased with boilup, approaching **100%** at higher values.
- Total energy consumption decreased from approximately **12.5 MW** to **11.2 MW** as boilup increased.

This demonstrates a clear engineering trade-off between product purity and recovery.

Increasing vapor generation improves ethanol recovery but reduces separation selectivity due to excessive vapor flow within the extractive column.

3. Effect of Column 2 Bottom Flow Rate

Bottom flow rate primarily controlled the solvent recovery section.

The study showed:

- Ethylene glycol recovery increased almost linearly with bottom flow.
- Recovery improved from approximately **68%** at **0.06 kg/s** to nearly **100%** at **0.09 kg/s**.
- Beyond **0.09 kg/s**, no meaningful improvement in solvent recovery was observed.

This indicates that **0.09 kg/s** is an appropriate operating point for maintaining solvent inventory while avoiding unnecessary recycle.

4. Effect of Column 2 Reflux Ratio

Column 2 reflux ratio mainly influenced energy consumption rather than separation performance.

The analysis showed:

- Increasing reflux ratio from **1.0** to **3.0** increased total energy consumption from approximately **8.6 MW** to over **12 MW**.
- Product recoveries changed only marginally.
- The highest overall Process Performance Index (**PPI = 0.951**) was achieved at a reflux ratio of **1.0**.

Therefore, excessive internal reflux increased utility consumption without providing significant process benefits.

5. Feed Temperature Behaviour

Feed temperature was found to be one of the most critical operating parameters.

The analysis identified two distinct operating regions.

Stable operating region (180–240°C)

Within this temperature range:

- Ethanol purity remained consistently high (approximately **97 wt%**).
- Ethanol recovery increased steadily with increasing temperature.

Critical operating limit (250°C)

At **250°C**, ethanol purity suddenly decreased from approximately **97 wt%** to **68.8 wt%**, while recovery remained high.

This sharp deterioration suggests a thermodynamic operating limit where excessive feed enthalpy disrupts the extractive separation mechanism.

Consequently, feed temperatures above approximately **240°C** should be avoided.

6. Interaction between Boilup Ratio and Feed Temperature

The combined sensitivity study demonstrated that process performance depends on the simultaneous adjustment of operating variables rather than independent optimization.

Moderate boilup ratios combined with feed temperatures between **200–220°C** consistently produced:

- High ethanol purity
- High ethanol recovery
- Reduced energy consumption
- High Process Performance Index

This confirms the existence of an optimal operating window rather than a single optimum variable.

7. Correlation Analysis

The Pearson correlation analysis quantified the influence of each process variable.

Major observations include:

- **Temperature showed the strongest correlation with ethanol recovery (r = 0.97)**, indicating that feed temperature is the dominant factor controlling recovery.
- **Bottom flow exhibited an almost perfect correlation with EG recovery (r ≈ 1.00)**, confirming that solvent recovery is governed primarily by Column 2 bottom flow.
- **Boilup ratio showed strong positive correlations with both ethanol purity (r = 0.79) and ethanol recovery (r = 0.80)**, highlighting its importance in Column 1 performance.
- **Reflux ratio displayed the strongest correlation with total energy consumption among the manipulated variables (r = 0.55)**, indicating that increasing reflux mainly increases utility requirements.
- Pressure exhibited only moderate correlations (approximately **0.45–0.50**) with most outputs, suggesting that pressure influences the process but is less significant than boilup, temperature, or bottom flow.

8. Variable Sensitivity Ranking

Based on the complete engineering analysis, the operating variables can be ranked according to their overall influence on process performance.

Rank	Operating Variable	Primary Influence
1	Feed Temperature	Ethanol Recovery & Process Stability
2	Boilup Ratio	Purity, Recovery and Energy
3	Bottom Flow Rate	Ethylene Glycol Recovery
4	Reflux Ratio	Total Energy Consumption
5	Column Pressure	Secondary Process Influence

9. Overall Engineering Interpretation

The complete sensitivity analysis demonstrates that the extractive distillation system possesses a relatively narrow operating window where product quality, solvent recovery, and energy consumption are simultaneously optimized.

The recommended operating region obtained from the engineering analysis is:

Parameter	Recommended Range
Column 1 Boilup Ratio	**0.44–0.47**
Column 2 Bottom Flow	**0.09 kg/s**
Column 2 Reflux Ratio	**1.0–1.5**
Feed Temperature	**220–240°C**
Column 2 Pressure	**≈1 bar**

Within this operating window, the process consistently achieved:

- Ethanol purity above **97 wt%**
- Ethanol recovery above **95%**
- Nearly complete ethylene glycol recovery

- Reduced energy consumption
- The highest Process Performance Index values

Final Conclusion

The engineering analysis demonstrates that **boilup ratio and feed temperature are the dominant variables controlling separation performance**, while **bottom flow rate governs solvent recovery** and **reflux ratio primarily affects energy consumption**. Rather than optimizing each variable independently, the study identifies an integrated operating window that balances product quality, recovery, and energy efficiency. These validated trends provide the engineering foundation for the machine-learning surrogate model developed in the subsequent stages of the project, enabling data-driven optimization of the extractive distillation process.